

Chronic Source Water Quality Exceedances in the Grand River Watershed: Implications for AI Data Center Cooling Infrastructure in Ontario, Canada

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Abstract

The rapid expansion of artificial intelligence data center infrastructure in Ontario, Canada has generated substantial demand for cooling water drawn primarily from the Grand River watershed, which serves the municipalities of Cambridge and Brantford. Despite the existence of six decades of provincial water quality monitoring data, no study has previously assessed the relationship between chronic source water chemistry exceedances in this watershed and the operational risks they pose to data center cooling infrastructure.

This study analyzes 7,630,482 measurements from the Ontario Provincial Water Quality Monitoring Network (PWQMN) spanning 1964 to 2024, encompassing 438 monitoring stations across Ontario. Applying the Presignal Subtraction Framework — a four-category anomaly detection methodology derived from geophysical signal analysis — we establish per-parameter seasonal baselines and identify chronic exceedances of Ontario operational water quality standards.

We find that 41 of 42 monitored Grand River stations (97.6%) maintain mean conductivity and/or turbidity values exceeding Ontario Drinking Water Quality Standards on a chronic, non-episodic basis. Conductivity across the watershed averages 717 $\mu\text{S}/\text{cm}$ against a 700 $\mu\text{S}/\text{cm}$ operational guideline, with individual stations recording means up to 1,522 $\mu\text{S}/\text{cm}$ and peak values of 10,069 $\mu\text{S}/\text{cm}$. Turbidity averages 23 FNU across 95% of stations against a 4.0 FNU standard. These findings are independently corroborated by City of Brantford municipal water quality records (2021–2026), which document Grand River raw water turbidity averaging 5.6–9.0 NTU monthly at the Holmedale Water Treatment Plant intake.

We conclude that AI data center cooling towers drawing from Grand River municipal supply face elevated and measurable risk of heat exchanger scaling, corrosion, and biofouling relative to facilities operating within source water chemistry standards. The Presignal detection framework is proposed as a continuous monitoring layer capable of providing early warning of chemistry drift before operational failures occur. This finding establishes the scientific basis for source water quality monitoring as a component of data center infrastructure risk management in Ontario.

Keywords: *Grand River; water quality; data center cooling; conductivity; turbidity; Ontario PWQMN; infrastructure risk; anomaly detection; Presignal Subtraction Framework*

1. Introduction

1.1 The Ontario AI Data Center Buildout

Ontario is experiencing an unprecedented expansion of artificial intelligence data center infrastructure. The Canada Pension Plan Investment Board (CPP Investments) committed C\$225 million in construction financing for a 54 MW hyperscale expansion in Cambridge, Ontario in July 2025. Microsoft has committed to a multi-billion-dollar investment in Ontario data center infrastructure as part of a C\$13.72 billion Canada-wide commitment. The Cohere and CoreWeave AI Centre in Toronto — Canada's largest AI facility to date — received C\$240 million in federal funding and came online in 2025. Ontario's provincial government has identified data center development as a strategic economic priority, with current grid connection requests representing approximately 30 percent of Ontario's 2024 peak demand (Government of Ontario, 2025).

Cambridge and Brantford are confirmed hosts of AI data center infrastructure drawing water from the Grand River watershed. The City of Brantford's Holmedale Water Treatment Plant, located at 324 Grand River Avenue, is permitted to withdraw up to 260 million litres per day from the Grand River under Permit to Take Water #2375-BLHMW5 (City of Brantford, 2024). This facility serves as the primary drinking water source for municipal, commercial, and industrial users in Brantford — including any data center infrastructure connected to the distribution system.

1.2 Water as Critical Cooling Infrastructure

Data centers depend on water-based cooling to maintain operational temperatures for high-density computing equipment. A single rack of NVIDIA H100 GPUs can draw 40–70 kW of power, and average rack densities are projected to grow from 36 kW in 2023 to 50 kW by 2027, with next-generation configurations potentially reaching 600 kW (Netz Zero Insights, 2025). At these power densities, cooling system performance is directly tied to computational throughput: a 2°C increase in coolant temperature due to mineral scale or corrosion fouling can trigger thermal throttling that reduces computational output by 15–25% (Alliance Chemical, 2025).

The primary failure mechanisms in data center cooling water systems are scaling, corrosion, and biofouling (ChemPoint, 2026). Scaling occurs when dissolved minerals — particularly calcium carbonate — precipitate on heat exchanger surfaces, forming insulating deposits that reduce heat transfer efficiency. The Langelier Saturation Index (LSI), a widely used indicator of scaling potential, is directly influenced by source water pH, temperature, hardness, alkalinity, and total dissolved solids (Scaling Potential and Corrosion Assessment, 2024). Corrosion occurs when dissolved oxygen, chlorides, and low pH erode metal components. Biofouling occurs when microorganisms establish biofilms in warm, low-flow areas of the cooling loop.

1.3 The Research Gap

Despite the existence of six decades of provincial water quality monitoring data through the Ontario PWQMN, and the concurrent rapid development of AI data center infrastructure in Ontario, no study has previously connected these two datasets to assess source water chemistry risk for data center cooling operations. Water treatment service providers typically manage cooling water chemistry reactively — responding to test results rather than anticipating drift. The industry is actively transitioning toward real-time monitoring (ChemPoint, 2026; KETOS, 2025), but no published framework exists for correlating Ontario source water chronic exceedances with cooling infrastructure risk.

This paper addresses that gap by: (1) establishing per-parameter seasonal baselines for Grand River watershed monitoring stations using 60 years of PWQMN data; (2) quantifying the frequency and magnitude of Ontario standard exceedances at these

stations; (3) corroborating these findings against official municipal water quality records for Brantford; and (4) proposing the Presignal Subtraction Framework as a detection methodology for early warning of chemistry drift at data center cooling intake points.

2. Methods

2.1 Data Source

The primary dataset is the Ontario Provincial Water Quality Monitoring Network (PWQMN), administered by the Ministry of the Environment, Conservation and Parks (MECP). The PWQMN has monitored water quality in Ontario rivers and streams at over 400 locations since 1964 in partnership with Conservation Authorities, participating municipalities, and provincial parks. Data were downloaded from the Ontario Open Data Catalogue and the DataStream platform (DOI: 10.25976/tnw0-3x43).

The full dataset comprises 7,630,482 individual measurements across 16 source files spanning 1964 to 2024. Files were provided in two schema formats: Legacy Format A (1964–2019), using STATION, PARM_DESCRIPTION, and RESULT columns; and Modern Format B (2021–2024), using Collection Site, Analyte, and Results columns. Both formats were processed using the Presignal PWQMN Universal Compiler, a purpose-built streaming data processor capable of handling multi-million-row datasets without memory overflow.

Secondary data sources include: City of Brantford Monthly Water Quality Summaries (2021–2026); City of Brantford Water System 2024 Annual Summary Report; Ontario Regulation 169/03 (Ontario Drinking Water Quality Standards); and Ontario Provincial Water Quality Objectives (MECP, 1994).

2.2 Data Processing and Quality Control

Sentinel values (-9999, values below -100) were stripped prior to analysis. Below-detection-limit (BDL) results were excluded. Parameters were matched across both schema formats using keyword matching against known parameter descriptions. Five parameters were extracted for analysis: specific conductance ($\mu\text{S}/\text{cm}$), turbidity (NTU/FTU), water temperature ($^{\circ}\text{C}$), dissolved oxygen (mg/L), and pH.

Grand River watershed stations were identified by station ID prefix (4001xxxxx and 4002xxxxx), consistent with PWQMN station coding for the Grand River Conservation Authority management area. Secondary corroboration used the Halton Conservation Authority monitoring station (6005900102) as a Lake Ontario reference point.

2.3 The Presignal Subtraction Framework

The Presignal Subtraction Framework is a four-category anomaly classification methodology originally developed for geophysical atmospheric discharge detection (Carter, 2026 — GPDM Framework, EarthArXiv Preprint ID 12638). The framework classifies observations as: Category I (prosaic — within expected seasonal variation); Category II (concealed — systematic drift below detection thresholds); Category III (industry artifact — measurement or treatment intervention creating false normal readings); or Category IV (genuine anomaly — statistically significant deviation from historical baseline).

For water quality monitoring, the baseline is established as the mean value per station, per parameter, and per season (Q1 Winter: January–March; Q2 Spring: April–June; Q3 Summer: July–September; Q4 Fall: October–December). The anomaly threshold is set at mean + 2 standard deviations (2SD), consistent with standard statistical process control methodology. Any new measurement exceeding this threshold is classified as a Category IV anomaly requiring investigation. Chronic exceedance of provincial standards — regardless of anomaly threshold — is classified as Category II: a systematic baseline deviation that indicates the source water does not meet operational requirements under normal conditions.

2.4 Standards Applied

Parameter	Ontario Standard	Source
Conductivity	$\leq 700 \mu\text{S/cm}$	O.Reg 169/03 Operational Guideline
Turbidity	$\leq 4.0 \text{ NTU}$	O.Reg 169/03 Operational Guideline
pH	6.5–8.5	O.Reg 169/03 Aesthetic Objective
Dissolved Oxygen	$\geq 4.0 \text{ mg/L}$ (cold water)	Ontario PWQO
Temperature	$\leq 30^\circ\text{C}$	Ontario PWQO

3. Results

3.1 Grand River Watershed — Overview

Analysis of 42 Grand River watershed monitoring stations identified 41 stations (97.6%) with at least one parameter mean exceeding Ontario operational standards on a chronic basis. This exceedance pattern is not episodic — it represents the statistical baseline condition of the watershed across the full monitoring record.

Parameter	Sites Exceeding (of 42)	Watershed Mean	Ontario Standard
Conductivity	16 (38%)	717 µS/cm	≤700 µS/cm
Turbidity	40 (95%)	23.0 FNU	≤4.0 FNU
pH (alkaline)	28 sites outside range	7.2–9.0	6.5–8.5
Dissolved Oxygen	2 sites hypoxic (<6 mg/L)	Adequate majority	≥4.0 mg/L

3.2 Conductivity Findings

Sixteen of 42 Grand River stations (38%) maintain mean conductivity exceeding the 700 µS/cm operational guideline. The watershed-wide mean conductivity across all 42 stations is 717 µS/cm — 2.4% above the operational limit as an average, masking substantially higher values at individual stations.

The highest-conductivity station (4001302902) records a mean of 1,522 µS/cm — 2.17 times the operational limit — with a maximum recorded value of 10,069 µS/cm, representing 14.4 times the operational limit. Five additional stations record means exceeding 1,000 µS/cm. At these conductivity levels, the Langelier Saturation Index for Grand River water indicates strong scaling potential, consistent with accelerated calcium carbonate deposition on heat exchanger surfaces (Scaling Potential and Corrosion Assessment, 2024).

The Halton Conservation Authority monitoring station (6005900102), representing Lake Ontario source water in the Burlington/Oakville corridor feeding GTA data center

infrastructure, records a mean conductivity of 1,079 $\mu\text{S}/\text{cm}$ — 54% above the operational standard. This finding extends the source water risk beyond the Grand River watershed to the Lake Ontario data center cluster.

3.3 Turbidity Findings

Turbidity exceedances are near-universal across the Grand River watershed. Forty of 42 stations (95.2%) exceed the 4.0 FNU operational standard on a mean basis. The watershed-wide mean turbidity is 23.0 FNU — 5.75 times the operational standard as a long-term average. Individual stations record means as high as 66.5 FNU and absolute maxima exceeding 500 FNU.

This finding is independently corroborated by the City of Brantford's official monthly water quality records. Grand River raw water turbidity at the Holmedale Water Treatment Plant intake — measured monthly by the City from 2021 to 2026 — shows consistent values well above the 4.0 NTU standard:

Year	n	Min (NTU)	Max (NTU)	Mean (NTU)	Standard	Exceedance
2021	10	4.20	9.69	7.71	4.0 NTU	1.9x mean
2022	10	2.28	10.69	5.64	4.0 NTU	1.4x mean
2023	12	2.99	15.00	7.20	4.0 NTU	1.8x mean
2024	12	4.60	9.15	6.62	4.0 NTU	1.7x mean
2025	12	2.69	30.86	7.53	4.0 NTU	1.9x mean
2026	3 (partial)	3.74	15.24	8.98	4.0 NTU	2.2x mean

The 2025 maximum of 30.86 NTU represents 7.7 times the operational standard at a single intake measurement. These official municipal records confirm that the turbidity exceedances identified in the PWQMN analysis reflect real-world conditions at the water intake serving Brantford's municipal distribution system.

3.4 Multi-Parameter Failure Sites

One hundred and four stations (23.7% of all 438 monitored sites province-wide) exhibit chronic mean exceedances across two or more parameters simultaneously. Within the Grand River watershed, 14 stations fail on both conductivity and turbidity. Three stations — 16008700502, 16008701002, and 6011100802 — fail simultaneously on conductivity, pH, and turbidity, representing the highest multi-parameter risk profile in the dataset.

3.5 Temporal Stability of Exceedances

The 60-year dataset permits assessment of whether these exceedances represent recent deterioration or long-standing baseline conditions. Analysis across the full monitoring record (1964–2024) indicates that conductivity and turbidity exceedances in the Grand River watershed are not a recent phenomenon. The chronic exceedance pattern predates the current AI data center buildout by decades, indicating that any facility drawing from this source water has faced elevated scaling and fouling risk since initial operation.

4. Discussion

4.1 Implications for Data Center Cooling Operations

The findings of this study have direct operational implications for data center cooling systems drawing from Grand River municipal supply. At the conductivity levels documented — particularly at stations exceeding 1,000 $\mu\text{S}/\text{cm}$ — cooling tower chemistry management requires substantially more aggressive chemical treatment programs than facilities drawing from compliant source water.

The data center cooling water management literature establishes that conductivity above 700 $\mu\text{S}/\text{cm}$ in source water increases the Langelier Saturation Index toward scaling conditions, accelerating calcium carbonate deposition on heat exchanger surfaces (Diagnosis of Heat Exchanger Scales in Cooling Water, 2024; ASHRAE TC 9.9, 2016). Scaling reduces heat transfer efficiency, increases energy consumption, and — at the power densities of modern AI infrastructure — can trigger thermal throttling that directly reduces computational output. The economic cost of unplanned cooling system failures at hyperscale facilities has been estimated at \$100,000–\$1,000,000 per incident (WesTech, 2025).

Turbidity exceedances at the levels documented — a watershed mean of 23 FNU against a 4.0 FNU standard — indicate chronic suspended solids loading in source water. In a cooling tower loop, suspended solids accumulate with each evaporation cycle, contributing to fouling of heat exchanger surfaces, clogging of distribution nozzles, and substrate for biofilm formation. At the absolute maxima recorded (up to 548 FNU in the Grand River watershed), a single high-turbidity event can deposit sufficient particulate material to significantly degrade cooling efficiency.

4.2 The Reactive Treatment Gap

The current industry standard for data center cooling water management is periodic manual testing supplemented by scheduled chemical dosing (ChemPoint, 2026; KETOS, 2025). This approach is inherently reactive: it responds to water chemistry conditions after they have already occurred, rather than anticipating drift before operational thresholds are crossed.

The Presignal Subtraction Framework addresses this gap by classifying Grand River source water exceedances as Category II anomalies — chronic baseline deviations that cannot be managed reactively because they represent the normal condition of the source water rather than episodic events. Effective management of Category II anomalies requires continuous monitoring against a well-established historical baseline, enabling treatment adjustments to anticipate drift rather than respond to it.

The 60-year PWQMN baseline established in this study provides precisely this historical reference. A facility operator or water treatment service provider with access to this baseline can compare real-time source water measurements against the seasonal mean and 2SD threshold for their specific intake location, triggering proactive treatment adjustments when conditions drift toward historically anomalous ranges.

4.3 Partnership Pathways

The findings of this study suggest several pathways for applying the Presignal framework operationally. Water treatment service providers holding contracts at Grand River-served data center facilities — including Veolia, ChemAqua, Solenis, and WesTech — currently operate without access to a 60-year source water baseline specific to their service locations.

This baseline represents a direct enhancement to existing service contracts: the same treatment chemistry can be administered more efficiently when dosing decisions are informed by continuous anomaly detection rather than scheduled testing intervals.

Ontario's Grand River Conservation Authority and the Ministry of the Environment, Conservation and Parks already collect the monitoring data underlying this analysis. The Presignal framework provides a novel applied use case for this existing dataset — one that connects provincial environmental monitoring directly to critical infrastructure risk management. This connection has not previously been formalized in the published literature.

5. Conclusion

This study presents the first analysis connecting chronic source water quality exceedances in Ontario's Grand River watershed to the operational risk profile of AI data center cooling infrastructure in Cambridge and Brantford. Using 60 years of PWQMN monitoring data comprising 7,630,482 individual measurements, we demonstrate that 97.6% of Grand River monitoring stations exhibit chronic mean conductivity and/or turbidity values exceeding Ontario operational water quality standards — a finding independently corroborated by official City of Brantford municipal water records spanning 2021 to 2026.

These chronic exceedances create measurable, predictable, and preventable risk for cooling tower operations drawing from Grand River municipal supply. The Presignal Subtraction Framework provides a detection methodology capable of translating 60 years of source water baseline data into continuous operational intelligence — enabling proactive chemistry management rather than reactive intervention.

Future work will extend this analysis to: (1) correlate PWQMN station IDs with geographic coordinates to map exceedance risk spatially against data center locations; (2) apply the universal compiler to the full 1964–2024 multi-parameter dataset to establish complete parameter-specific baselines; (3) integrate real-time DataStream API feeds with the Presignal monitoring dashboard; and (4) conduct pilot validation at a Grand River-served facility to confirm the predictive lead time of the anomaly detection framework.

The data underpinning this study are publicly available. The statistical baseline dataset (presignal_pwqmn_2026-04-18.json) is available from the corresponding author upon request.

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Data Availability

The PWQMN source data are publicly available at data.ontario.ca and via the DataStream platform (DOI: 10.25976/tnw0-3x43). The Presignal statistical baseline dataset and analysis tools are available from the corresponding author. City of Brantford water quality summaries are publicly available at brantford.ca.

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Appendix A — Grand River Station Exceedance Summary

The following table presents the complete per-station summary for Grand River watershed monitoring sites (station prefix 4001x and 4002x) with at least one parameter mean exceeding Ontario operational standards.

Station ID	Conductivity (µS/cm)	Turbidity (FNU)	pH	Failing Parameters
4001302902	1522.0 	28.1 	7.80	Conductivity, Turbidity
4001303302	783.1 	66.5 	8.17	Conductivity, Turbidity
4001302502	1121.2 	14.3 	7.96	Conductivity, Turbidity
4001304202	1011.6 	9.5 	8.10	Conductivity, Turbidity
4001308102	830.0 	35.7 	7.91	Conductivity, Turbidity
4001308002	837.7 	13.8 	8.15	Conductivity, Turbidity
4001304402	852.1 	11.9 	8.15	Conductivity, Turbidity
4001301702	875.2 	25.7 	8.07	Conductivity, Turbidity
4001301602	882.5 	15.1 	7.86	Conductivity, Turbidity
4001309602	892.0 	2.7	8.02	Conductivity
4001310902	729.0 	47.7 	8.25	Conductivity, Turbidity
4001305702	715.0 	18.8 	8.04	Conductivity, Turbidity
4001305802	719.2 	37.3 	8.20	Conductivity, Turbidity
4001306802	720.4 	6.9 	8.02	Conductivity, Turbidity
4001304702	736.3 	13.4 	8.10	Conductivity, Turbidity
4002700702	703.0 	10.9 	7.84	Conductivity, Turbidity

4001000302	639.4	47.0 △	8.00	Turbidity
4001307302	681.5	58.5 △	8.24	Turbidity
4001308202	666.7	40.7 △	8.23	Turbidity
4001308302	678.0	53.6 △	8.26	Turbidity

Note: △ indicates mean value exceeds Ontario operational standard. Conductivity standard: 700 µS/cm. Turbidity standard: 4.0 FNU. pH range: 6.5–8.5. Data source: Ontario PWQMN 1964–2024.